

Semester-wise Pattern for Students Admitted to B.E. Environmental and Sustainability Engineering Programme								
Year	First Semester			U	Second Semester			U
I	BITS	F103	Introduction to Engineering Design and Prototyping	4	BITS	F111	Thermodynamics	3
	BIO	F101	Introduction to Biological Sciences	3	BITS	F112	Technical Report Writing	2
	CHEM	F101	Fundamentals of Chemistry	3	CS	F111	Computer Programming	4
	MATH	F101	Multivariable Calculus	3	MATH	F113	Probability and Statistics	3
	OR		OR					
	BITS	F113	General Mathematics I	3				
	PHY	F101	Oscillations and Waves	3	MATH	F102	Linear Algebra and Complex Variables	3
	OR		OR		OR		OR	
	PHY	F102	Introduction to Oscillations and Waves	3	BITS	F114	General Mathematics II	3
	BITS	F101	Social Conduct	1	EEE	F111	Electrical Sciences	3
BITS	F102	Innovation and Design Thinking	1	BITS	K101	Physical Well-being and Creativity	1*	
				20				18
II	MATH	F211	Mathematics III	3	ECON	F211	Principles of Economics	3
	OR		OR		OR		OR	
	BITS	F218	General Mathematics		MGTS	F211	Principles of Management	
	CHE	F211	Chemical Process Calculations	3	ENVS	F221	Water and Wastewater Treatment and Management	3
	ENVS	F211	Hydraulics & Fluid Mechanics	3	ENVS	F222	Transport Phenomena for Environmental Systems	3
	ENVS	F212	Environmental Chemistry and Biotechnology	3	ENVS	F223	Material Science and Engineering for Sustainability	3
	BITS	F240	Introduction to Environmental & Sustainable Systems Engineering	3	ENVS	F224	Sampling and Analytical Techniques for Pollutants	3
	BITS	F225	Environmental Studies	3			Humanities Elective	3
		Humanities Elective	3					
			21				18	
Summer BITS F221 Practice School – I								
III			Open/Humanities Electives	3 to 6			Open/Humanities Electives	3 to 6
	ENVS	F311	Environmental Modelling	3	ENVS	F321	Sustainability Policies and Measures	3
	ENVS	F312	Air Pollution Control	3	ENVS	F322	Solid and Hazardous Waste Management	3
	ENVS	F313	Digital Sustainability	3	ENVS	F323	Sustainable Urban Design and Smart Cities	3
	ENVS	F314	Environmental Engineering Lab	3	ENVS	F324	Environmental Economics and Governance	3
		Discipline Electives	3			Discipline Elective	3	
			18/21				18/21	
IV			Open/Humanities Electives	6 to 11	BITS	F412	Practice School-II	20
			Discipline Electives	6			or	or
					BITS	F421T	Thesis	16
					or	F425T	Thesis	18-20
					or	F424T	Thesis (9) and Electives (6 to 9)	15-18
				12/17				15/20

Discipline Core - 48 Units (16 Courses)

Discipline Electives - 12 Units (4 Courses)